

PHK Early Warning Dashboard

Technical Notes

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1 Methodology for Provincial PHK Projection

1.1 Objective

The objective of the model is to project the monthly number of layoffs (PHK) at the provincial level as an early warning indicator of labour market deterioration. The model combines structural characteristics of provincial labour markets with high-frequency macroeconomic indicators to capture both long-term labour market vulnerability and short-term economic shocks.

1.2 Econometric Approach

- The dependent variable is the monthly number of reported layoffs (PHK) by province, which is a non-negative count variable with a substantial number of zero observations.
- A two-way Fixed Effects (FE) Poisson regression is employed:

$$E(PHK_{it}|X_{it}) = \exp(\beta X_{it} + \alpha_i + \gamma_t),$$

where:

- PHK_{it} denotes the number of layoffs in province i during month t ;
 - α_i represents province fixed effects;
 - γ_t represents month fixed effects;
 - X_{it} contains structural and macroeconomic variables.
- Robust standard errors are clustered at the provincial level to account for within-province serial correlation.

1.3 Rationale for Model Selection

- Provincial labour markets differ systematically in industrial composition, labour institutions, economic structure, and reporting practices. Since these unobserved characteristics are likely correlated with macroeconomic variables, the assumptions underlying Random Effects estimation are unlikely to hold. Fixed Effects therefore provide more consistent estimates by controlling for time-invariant provincial heterogeneity (Hausman, 1978; Wooldridge, 2010).
- Compared with time-series approaches such as Vector Autoregression (VAR), panel Fixed Effects models exploit both cross-sectional and time-series variation while accommodating heterogeneous provincial characteristics and relatively short monthly panels (Baltagi, 2021; Sims, 1980).
- Poisson regression is preferred because the dependent variable represents monthly counts of layoffs rather than continuous outcomes or proportions. Compared with OLS, Poisson guarantees non-negative predictions and naturally accommodates zero observations. Compared with Fractional Logit, which is designed for fractional outcomes bounded between zero and one, Poisson is more appropriate for modelling event counts. Poisson Fixed Effects estimators remain consistent under relatively weak assumptions on the conditional mean and are robust to many forms of

heteroskedasticity (Hausman et al., 1984; Santos Silva and Tenreyro, 2006; Cameron and Trivedi, 2013; Papke and Wooldridge, 1996).

1.4 Variables

Dependent Variable

- Monthly provincial PHK flow (number of reported layoffs).

Structural Variables (Annual)

- Gross Regional Domestic Product (GRDP, log)
- Manufacturing share of GRDP
- Manufacturing employment share
- Formal employment share
- Provincial minimum wage (log)

Macroeconomic Trigger Variables (Monthly)

- *Global*: Brent crude oil price.
- *National*: BI policy rate, IDR/USD exchange rate, Manufacturing PMI, Producer Price Index (IHPB).
- *Provincial*: Consumer Price Index (CPI), Non-Performing Loans (NPL), export value, and import value.

1.5 Lag Specification and Model Selection

- Macroeconomic trigger variables are evaluated using four alternative lag structures (1, 3, 6, and 12 months) to capture delayed responses of firms to economic shocks, while structural variables enter contemporaneously.
- (TBD) Candidate models are estimated using the training sample (2023–2025) and evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) on the validation sample.
- (TBD) The specification with the lowest prediction error is selected as the preferred forecasting model and subsequently re-estimated using the full estimation sample before generating provincial and national PHK projections.

2 Results

2.1 Single-Lag Specifications

Table 1: Poisson regression estimates for provincial layoff projections

		(1) Lag 1	(2) Lag 3	(3) Lag 6	(4) Lag 12
Structure Variables					
	GRDP (log)	-10.044 (8.856)	-0.034 (7.566)	-15.277 (12.476)	-19.247* (9.665)
	Manufacturing Share in GRDP (%)	0.125 (0.139)	0.010 (0.144)	0.159 (0.118)	0.123 (0.104)
	Manufacturing Workers (%)	-0.170 (0.137)	-0.132 (0.133)	-0.150 (0.149)	-0.194 (0.212)
	Formal Workers (%)	-0.170 (0.101)	-0.158 (0.097)	-0.151 (0.108)	-0.142 (0.114)
	Minimum Wage (log)	9.606 (7.183)	13.083 (8.493)	-2.816 (10.029)	13.813 (13.158)
Macroeconomic Trigger Variables					
% yoy	Brent Oil Price (log)	3.003 (1.901)	0.293 (1.001)	-1.534 (1.144)	4.004*** (0.965)
	BI Rate	0.138 (0.383)	0.808*** (0.219)	0.195 (0.142)	0.839* (0.370)
rata-rata SD bulanan	IDR/USD Exchange Rate (log)	-9.390 (7.045)	-0.047 (2.677)	0.209 (3.274)	-13.665* (5.590)
	PMI Manufacturing	-0.114* (0.058)	-0.019 (0.069)	-0.036 (0.038)	-0.043 (0.109)
% yoy	Wholesale Price Index	0.530** (0.193)	-0.071 (0.254)	0.724*** (0.173)	0.325 (0.209)
% yoy	Consumer Price Index	-0.170 (0.137)	-0.074 (0.140)	-0.389* (0.175)	-0.160 (0.149)
rasio NPL/total credit %yoy	Non-Performing Loan (log)	0.095 (0.365)	-0.248 (0.455)	-0.005 (0.631)	-0.871 (0.707)
%yoy	Export (log)	-0.290 (0.235)	-0.932* (0.471)	0.065 (0.318)	0.157 (0.193)
%yoy	Import (log)	-0.210* (0.083)	-0.107 (0.102)	-0.003 (0.082)	0.152* (0.075)
Observations		977	975	972	964
DV Mean		233.99	233.79	233.99	236.71
Pseudo R ²		0.582	0.581	0.584	0.591
Region FE		Province	Province	Province	Province
Time FE		Month	Month	Month	Month

Notes: The dependent variable is the monthly number of layoffs (PHK) by province. Robust standard errors clustered at the province level are reported in parentheses. Columns (1)-(4) report Poisson regression estimates using macroeconomic trigger variables lagged by 1, 3, 6, and 12 months, respectively. Annual structural variables are included contemporaneously in all specifications. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

2.2 Single-Lag Specifications - Robustness Check 1

Table 2: Poisson regression estimates for provincial layoff projections

	(1) Lag 1	(2) Lag 3	(3) Lag 6	(4) Lag 12
Structure Variables				
log_pdrb_y	-10.417 (9.177)	-0.084 (7.575)	-15.291 (12.498)	-18.707* (9.494)
manuf_pdrb_share_y	0.118 (0.138)	0.006 (0.144)	0.151 (0.125)	0.141 (0.109)
manuf_emp_share_y	-0.145 (0.161)	-0.116 (0.187)	-0.117 (0.198)	-0.335 (0.324)
formal_lab_share_y	-0.175 (0.103)	-0.160 (0.101)	-0.155 (0.108)	-0.136 (0.112)
log_wage_ump_y	9.274 (7.452)	12.916 (8.891)	-3.181 (10.648)	14.767 (13.646)
log_wage_avg_y	1.225 (2.863)	0.560 (2.816)	0.967 (2.877)	-2.822 (2.892)
Macroeconomic Trigger Variables				
L_log_price_brent	3.015 (1.885)	0.280 (1.013)	-1.542 (1.152)	4.068*** (0.944)
L_bi_rate_nat	0.131 (0.370)	0.808*** (0.220)	0.199 (0.145)	0.873* (0.363)
L_log_fx_idr_usd	-9.371 (6.993)	-0.067 (2.644)	0.133 (3.368)	-14.167** (5.397)
L_pmi_manuf_nat	-0.113* (0.057)	-0.019 (0.069)	-0.038 (0.037)	-0.041 (0.108)
L_ihpb_nat	0.529** (0.190)	-0.074 (0.246)	0.722*** (0.171)	0.327 (0.211)
L_cpi_index	-0.173 (0.137)	-0.074 (0.139)	-0.391* (0.175)	-0.167 (0.157)
L_log_npl	0.257 (0.638)	-0.187 (0.656)	0.085 (0.856)	-1.020 (0.751)
L_log_export	-0.288 (0.234)	-0.933* (0.469)	0.064 (0.316)	0.160 (0.193)
L_log_import	-0.205* (0.082)	-0.106 (0.101)	-0.005 (0.082)	0.158* (0.077)
Observations	977	975	972	964
DV Mean	233.99	233.79	233.99	236.71
Pseudo R ²	0.583	0.581	0.584	0.592
Region FE	Province	Province	Province	Province
Time FE	Month	Month	Month	Month

Notes: The dependent variable is the monthly number of layoffs (PHK) by province. Robust standard errors clustered at the province level are reported in parentheses. Columns (1)-(4) report Poisson regression estimates using macroeconomic trigger variables lagged by 1, 3, 6, and 12 months, respectively. Annual structural variables are included contemporaneously in all specifications. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

2.3 Single-Lag Specifications - Robustness Check 2

Table 3: Poisson regression estimates for provincial layoff projections

	(1) Lag 1	(2) Lag 3	(3) Lag 6	(4) Lag 12
Structure Variables				
log_pdrb_y	-10.941 (8.689)	-0.651 (6.915)	-15.639 (11.663)	-18.901* (9.322)
manuf_pdrb_share_y	0.136 (0.142)	0.025 (0.144)	0.165 (0.113)	0.147 (0.120)
manuf_emp_share_y	-0.133 (0.159)	-0.103 (0.184)	-0.107 (0.202)	-0.329 (0.320)
agri_emp_share_y	0.067 (0.120)	0.069 (0.109)	0.050 (0.117)	0.023 (0.117)
formal_lab_share_y	-0.168 (0.101)	-0.154 (0.102)	-0.151 (0.107)	-0.133 (0.108)
log_wage_ump_y	9.846 (7.800)	13.494 (9.100)	-2.868 (10.527)	14.893 (13.804)
log_wage_avg_y	1.402 (2.771)	0.789 (2.686)	1.130 (2.747)	-2.757 (2.772)
Macroeconomic Trigger Variables				
L_log_price_brent	3.028 (1.894)	0.272 (0.995)	-1.537 (1.155)	4.063*** (0.944)
L_bi_rate_nat	0.124 (0.366)	0.804*** (0.216)	0.199 (0.141)	0.870* (0.362)
L_log_fx_idr_usd	-9.274 (6.994)	-0.060 (2.646)	0.090 (3.413)	-14.123*** (5.381)
L_pmi_manuf_nat	-0.113* (0.057)	-0.018 (0.068)	-0.039 (0.038)	-0.041 (0.107)
L_ihpb_nat	0.523** (0.191)	-0.079 (0.244)	0.720*** (0.174)	0.323 (0.212)
L_cpi_index	-0.171 (0.137)	-0.073 (0.139)	-0.391* (0.173)	-0.162 (0.149)
L_log_npl	0.274 (0.642)	-0.141 (0.640)	0.137 (0.849)	-1.012 (0.740)
L_log_export	-0.309 (0.249)	-0.956* (0.485)	0.054 (0.311)	0.162 (0.195)
L_log_import	-0.204* (0.084)	-0.106 (0.101)	-0.009 (0.084)	0.157* (0.079)
Observations	977	975	972	964
DV Mean	234	234	234	237
Pseudo R ²	0.583	0.582	0.584	0.592
Region FE	Province	Province	Province	Province
Time FE	Month	Month	Month	Month

Notes: The dependent variable is the monthly number of layoffs (PHK) by province. Robust standard errors clustered at the province level are reported in parentheses. Columns (1)-(4) report Poisson regression estimates using macroeconomic trigger variables lagged by 1, 3, 6, and 12 months, respectively. Annual structural variables are included contemporaneously in all specifications. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

2.4 Cumulative Lag Specifications

Table 4: Comparison of alternative lag specifications

	(1) Lag 1	(2) Lag 1+3	(3) Lag 1+3+6	(4) Lag 1+3+6+12
Structure Variables				
log_pdrb_y	-10.044	-6.462	-5.074	2.852
manuf_pdrb_share_y	0.125	0.098	0.084	-0.027
manuf_emp_share_y	-0.170	-0.218	-0.238	-0.238
formal_lab_share_y	-0.170	-0.154	-0.129	-0.152
log_wage_ump_y	9.606	4.451	-5.761	13.044
Macroeconomic Trigger Variables				
L1_log_price_brent	3.003	0.657	3.365	1.629
L1_bi_rate_nat	0.138	0.137	1.117	0.500
L1_log_fx_idr_usd	-9.390	7.774	19.332	32.150**
L1_pmi_manuf_nat	-0.114*	-0.037	-0.151	-0.332**
L1_ihpb_nat	0.530**	1.077***	-0.298	-0.802
L1_cpi_index	-0.170	-0.008	0.056	0.132
L1_log_npl	0.095	0.515	0.490	0.868
L1_log_export	-0.290	-0.079	-0.147	-0.291
L1_log_import	-0.210*	-0.243*	-0.180	-0.117
L3_log_price_brent		-0.501	4.557***	5.644**
L3_bi_rate_nat		0.968	-3.488**	-4.852**
L3_log_fx_idr_usd		0.159	14.551	7.181
L3_pmi_manuf_nat		-0.141*	-0.139	-0.196
L3_ihpb_nat		-0.894**	-0.937**	0.176
L3_cpi_index		-0.159	-0.147	-0.125
L3_log_npl		-0.757	-0.935	-1.099
L3_log_export		-0.766	-0.825	-0.942
L3_log_import		-0.089	-0.130	-0.166
L6_log_price_brent			-4.787***	2.147
L6_bi_rate_nat			2.568***	4.797*
L6_log_fx_idr_usd			22.642	2.793
L6_pmi_manuf_nat			0.063	-0.042
L6_ihpb_nat			0.826	-0.351
L6_cpi_index			-0.322	-0.411
L6_log_npl			0.207	0.730
L6_log_export			0.055	-0.067
L6_log_import			-0.001	-0.079
L12_log_price_brent				5.418*
L12_bi_rate_nat				1.021
L12_log_fx_idr_usd				-39.013**
L12_pmi_manuf_nat				-0.115
L12_ihpb_nat				0.421
L12_cpi_index				-0.293
L12_log_npl				-0.897
L12_log_export				0.054
L12_log_import				0.134
Observations	977	951	928	898
Mean monthly PHK	233.99	238.83	243.23	249.57
Pseudo R ²	0.582	0.600	0.623	0.638
Province FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes

2.5 Discussion of Preliminary Findings

Table 5 summarizes the main findings from the cumulative lag specifications. Since multiple lagged values of the same variable are estimated simultaneously, the emphasis is placed on the timing and consistency of the effects rather than the magnitude of individual coefficients, which may be affected by multicollinearity across lags.

Table 5: Summary of Main Findings from Alternative Lag Specifications

Variable	Main Timing	Interpretation
Exchange Rate (IDR/USD)	1 and 12 months	Immediate increase in production costs followed by longer-term firm restructuring.
Manufacturing PMI	1 month	Early business cycle indicator reflecting changes in production and new orders.
BI Policy Rate	3–6 months	Delayed monetary transmission through financing costs and investment decisions.
Brent Oil Price	3–12 months	Gradual adjustment to higher production and transportation costs.
Producer Price Index (IHPB)	1 month	Immediate producer cost pressures affecting firms' operations.
Import Value	1 month	Proxy for production activity and availability of intermediate inputs.

Interpretation

- **Exchange rate.** The exchange rate emerges as one of the strongest leading indicators of layoffs, with significant effects in both the short run (1 month) and longer horizon (12 months). A depreciation of the Rupiah raises imported input costs immediately, while persistent exchange rate pressures may lead firms to undertake more permanent employment adjustments. Similar employment responses to exchange rate movements are documented by [Campa and Goldberg \(2001\)](#) and [Goldberg and Tracy \(2003\)](#).
- **Manufacturing PMI.** The Manufacturing PMI is significant only at the one-month lag, suggesting that firms adjust employment rapidly following changes in business sentiment, production, and new orders. This finding is consistent with the OECD's characterization of PMI as one of the earliest indicators of business-cycle turning points ([OECD, 2024](#)).
- **BI Policy Rate.** The BI policy rate becomes significant after approximately three to six months, consistent with the delayed transmission of monetary policy through financing costs, investment decisions, and labour demand. This timing agrees with the monetary transmission literature ([Bernanke and Gertler, 1995](#); [Christiano et al., 1999](#)).
- **Brent Oil Price.** Oil price shocks exhibit medium- to longer-horizon effects (3–12 months), indicating that firms initially absorb higher energy costs before gradually adjusting production and employment. Similar delayed responses are documented by [Hamilton \(1983\)](#) and [Kilian \(2009\)](#).
- **Producer Price Index (IHPB).** Producer prices primarily affect layoffs in the short run, reflecting immediate increases in production costs. The absence of persistent effects suggests that

firms either absorb or gradually pass through cost shocks over time, consistent with labour demand adjustment theory ([Hamermesh, 1993](#)).

- **Import Value.** Higher import values are associated with lower subsequent layoffs, likely reflecting stronger manufacturing activity and greater availability of intermediate inputs, which support firms' production capacity ([Hamermesh, 1993](#)).

Overall, the results suggest that provincial layoffs are influenced by both *immediate business-cycle indicators* (PMI, IHPB, imports), *medium-term policy transmission* (BI policy rate), and *longer-term structural adjustment* (exchange rate and oil prices). This temporal pattern is broadly consistent with labour demand theory and empirical evidence on employment adjustment over the business cycle [Hamermesh, 1993](#); [Davis and Haltiwanger, 1992](#).

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